Lección 10

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April 20, 2026

Plan de trabajo

Formulación Lagrangiana

—Aplicaciones:

*Observadores no inerciales

*Movimiento de Sólido rígido

▶ Nuestro Syllabus:

Nuestro Syllabus:

FIS-310:

UNIDAD 2. Formulación Lagrangiana

2.1. Variables Generalizadas

2.2. Clasificación de los Sistemas Mecánicos

2.3. Elementos de Cálculo Variacional

2.4. Ecuaciones de Lagrange

2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales

FIS-345:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos

- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

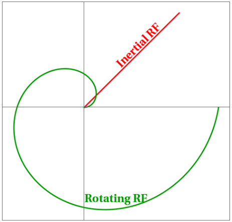
Unidad II: Algunas Aplicaciones

- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

Equations of motion in non-inertial system

$$L = \frac{m}{2}\left(v^2 + \underbrace{\left(\vec{\Omega} \times \vec{r}\right)^2}_{\text{centrifugal force}} + \underbrace{V^2(t)}_{\text{doesn't change EOM}} + \underbrace{2\vec{\boldsymbol{v}} \cdot \left(\vec{\Omega} \times \vec{r}\right)}_{\text{Coriolis}} - \underbrace{2\vec{\boldsymbol{r}} \cdot \frac{d\vec{\boldsymbol{V}}}{dt}}_{\text{inertia}} \right) - U(r)$$

$$\begin{aligned}
\frac{\partial L}{\partial \dot{r}_a} &\equiv \frac{\partial L}{\partial v_a} = m v_a + m\left(\vec{\Omega} \times \vec{r}\right)_a, \\
\frac{\partial L}{\partial r_a} &= -m\left[\vec{\Omega} \times [\vec{\Omega} \times \vec{r}]\right]_a + m\left[\vec{\boldsymbol{v}} \times \vec{\Omega}\right]_a - m\left(\frac{d\boldsymbol{V}}{dt}\right)_a - \frac{\partial U}{\partial r_a} \\
\Rightarrow m\dot{\boldsymbol{v}} &= -\underbrace{m\dot{\boldsymbol{\Omega}} \times \boldsymbol{r}}_{\text{Euler force}} - \underbrace{m\boldsymbol{\Omega} \times [\boldsymbol{\Omega} \times \boldsymbol{r}]}_{\text{centrifugal}} + 2\underbrace{m\boldsymbol{v} \times \boldsymbol{\Omega}}_{\text{Coriolis}} - \underbrace{m\frac{d\boldsymbol{V}}{dt}}_{\text{inertia}} - \frac{\partial U}{\partial \boldsymbol{r}}
\end{aligned}$$



Fictitious forces⇔equivalent results in different frames

Projectile launched from the center of disk:

–Inertial (lab frame): trajectory=is a straight line (red curve).

– Rotating frame: trajectory is a spiral (green curve) due to fictitious forces.

Control question

Assume that You analyze motion of the body in the laboratory frame, and would like to take into account that it is NOT inertial due to the rotation of the Earth. What will be the direction of the centrifugal and Coriolis forces?

Motion in rotating reference frame

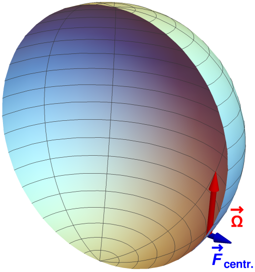
►Let's start with analysis of direction of the forces.
▷The centrifug. force does not depend on velocity of the bus nor its direction of motion

$$\boldsymbol{F}_{\text{centrif}} = -m\vec{\Omega} \times [\vec{\Omega} \times \vec{r}] = m\Omega^2 \boldsymbol{r}_\perp$$

►The vector $\boldsymbol{r}_\perp$ in the local frame may be decomposed into two parts:

▷ Vertical component

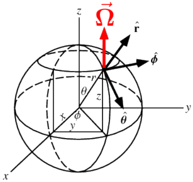
▷ Component pointing to the north (in southern hemisphere)



$\boldsymbol{F}_{\text{centrif.}}$ in local frame

$$\vec{\boldsymbol{F}} = -\underbrace{m\,\boldsymbol{\Omega}\times[\boldsymbol{\Omega}\times\boldsymbol{r}]}_{\text{centrifugial}} + \underbrace{2m\dot{\boldsymbol{r}}\times\boldsymbol{\Omega}}_{\text{Coriolis}}$$

▷Assume z -axis points to the North



▷Angular velocity:

$$\begin{aligned}\boldsymbol{\Omega} &= \Omega_0\hat{z}\\ \Omega_0 &= \left(\frac{2\pi}{T}\right),\\ T &= 24\times 60\times 60\\ &= 86\,400\ [\mathrm{sec}]\end{aligned}$$

► Relation of latitude Θ with polar angle θ:

$$\theta = \frac{\pi}{2} - \Theta$$

►In this problem easier to work in spherical coordinates

▷Eastward motion: $\dot{\boldsymbol{r}} \sim \hat{\phi}$

▷Southward motion: $\dot{\boldsymbol{r}} \sim \hat{\boldsymbol{\theta}}$

▷ $\hat{z} = \hat{\boldsymbol{r}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta$

−Decomposition: $\boldsymbol{r} = R_0\hat{\boldsymbol{r}}$ where R_0 is the radius of the Earth

$$\boldsymbol{\Omega}\times[\boldsymbol{\Omega}\times\boldsymbol{r}] = \boldsymbol{\Omega}(\Omega\cdot r) - \boldsymbol{r}\,\Omega^2 = \Omega_0^2 R_0\left((\hat{\boldsymbol{r}}\cos\theta - \hat{\boldsymbol{\theta}}\sin\theta)\cos\theta - \hat{\boldsymbol{r}}\right)$$

$$= -\Omega_0^2 R_0\left(\hat{\boldsymbol{r}}\,\sin^2\theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\theta}{2}\right) = -\Omega_0^2 R_0\left(\hat{\boldsymbol{r}}\,\cos^2\Theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\Theta}{2}\right)$$

$$\Rightarrow \boldsymbol{F}_{\text{centrifugial}} = -m\,\boldsymbol{\Omega}\times[\boldsymbol{\Omega}\times\boldsymbol{r}] = m\Omega_0^2 R_0\left(\hat{\boldsymbol{r}}\,\cos^2\Theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\Theta}{2}\right)$$

▷Centrifugal force does not depend on velocity at all

$$\Rightarrow \boldsymbol{F}_{\text{centrifugial}} = -m\,\boldsymbol{\Omega}\times[\boldsymbol{\Omega}\times\boldsymbol{r}] = m\Omega_0^2 R_0\left(\hat{\boldsymbol{r}}\,\cos^2\Theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\Theta}{2}\right)$$

- For typical lab experiments we have typical distances $\ll R_0$ (Earth's radius), so

$$\Theta \approx \text{const} \Rightarrow \boldsymbol{F}_{\text{centrifugial}} \approx \text{const}$$

- Note that centrifugial term simply might be absorbed into redefinition of $\boldsymbol{g}$:

$$\boldsymbol{P} + \boldsymbol{F}_{\text{centrifugial}} = m\left[-g\,\hat{\boldsymbol{r}} + \Omega_0^2 R_0\left(\hat{\boldsymbol{r}}\,\cos^2\Theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\Theta}{2}\right)\right] = -m\boldsymbol{g}_{\text{eff}}$$

$$\boldsymbol{g}_{\text{eff}} = -g\,\hat{\boldsymbol{r}} + \Omega_0^2 R_0\left(\hat{\boldsymbol{r}}\,\cos^2\Theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\Theta}{2}\right)$$

where $g \approx 9.81\,m/s^2$, $\Omega_0^2 R_0 \approx 0.03\,m/s^2$

More formal approach:

Decompose radius-vector (position) $\boldsymbol{r}$ as $\boldsymbol{r} \approx R_0\hat{\boldsymbol{r}} + \delta\boldsymbol{r}$

$$\Rightarrow \boldsymbol{F}_{\text{centrifugial}} = -m\,\boldsymbol{\Omega}\times[\boldsymbol{\Omega}\times\boldsymbol{r}] \approx m\Omega_0^2 R_0\left(\hat{\boldsymbol{r}}\,\cos^2\Theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\Theta}{2}\right) - m\,\boldsymbol{\Omega}\times[\boldsymbol{\Omega}\times\delta\boldsymbol{r}]$$

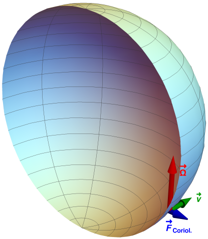
- In view of smallness of Ω, usually the last term is negligibly small and may be disregarded completely.

► For Coriolis force we have

$$\boldsymbol{F}_{\text{Coriolis}} = 2m\boldsymbol{v} \times \boldsymbol{\Omega}$$

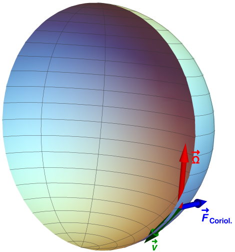
so the result depends on the direction of motion of the object.

Motion to the east:



► The Coriolis force has same direction as $\boldsymbol{F}_{\text{centrif}}$, might be decomposed in local frame into two components pointing in vertical and northern directions.

Motion to the south:



► The Coriolis force is directed to the east. Force depends on $\boldsymbol{v}$: for motion to the north, the force $\boldsymbol{F}_{\text{Coriolis}}$ would change its sign and point to the west, etc.

F_{Coriolis} vs F_{centrif}

Who is larger?

–Depends on the context (velocities, angular velocity of the non-inertial frame)

–For estimates related to rotation of the Earth (e.g. Foucault pendulum, etc), Ω is small parameter,

$$\boldsymbol{F}_{\text{Coriolis}} = 2m\boldsymbol{v} \times \boldsymbol{\Omega}$$

$$\begin{aligned}
\boldsymbol{P} + \boldsymbol{F}_{\text{centrifugial}} &= m\left[-g\,\hat{\boldsymbol{r}} + \Omega_0^2 R_0 \left(\hat{\boldsymbol{r}}\cos^2\Theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\Theta}{2}\right)\right] \\
&= -m\boldsymbol{g}_{\text{eff}} - m\,\boldsymbol{\Omega} \times [\boldsymbol{\Omega} \times \delta\boldsymbol{r}] \\
\boldsymbol{g}_{\text{eff}} &= -g\,\hat{\boldsymbol{r}} + \Omega_0^2 R_0 \left(\hat{\boldsymbol{r}}\cos^2\Theta + \hat{\boldsymbol{\theta}}\frac{\sin 2\Theta}{2}\right)
\end{aligned}$$

the red-colored term is $\sim \Omega^2$ and thus can be disregarded compared to Coriolis ($\sim \Omega$)

$\Rightarrow$We'll focus on Coriolis in what follows and disregard centrifugal force

When we solve problems in non-inertial reference frame, use the standard algorithm:

—Write out all forces

—drop terms which are negligible
(e.g., the Earth rotation is slow$\Rightarrow\Omega$ can be used as a small parameter)

—try to solve the system of ODE's

—sometimes these forces may lead to new qualitative effects ...

* Expectation: fictitious forces should rotate trajectores ...

Please try to recall from undergraduate courses examples of <u>qualitative</u> effects due to rotation of the Earth

Problem 1

Assume that You study the scattering of an electron in the central potential field $U(r)$ [$for\ example,\ U(r) = -\kappa/r$]. Analyze how the rotation of the Earth will affect its trajectory

Problem 1 (solution)

Assume $\boldsymbol{\Omega} = \Omega\, \boldsymbol{n} = \mathrm{const}$ is the vector of angular velocity
Equations of motion in non-inertial frame:

$$m\, \ddot{\boldsymbol{r}} = -2m\boldsymbol{\Omega} \times \boldsymbol{v} - m\,\boldsymbol{\Omega} \times [\boldsymbol{\Omega} \times \boldsymbol{r}] - \boldsymbol{\nabla} U, \qquad \boldsymbol{\nabla} U = \frac{\partial U}{\partial r} \hat{\boldsymbol{r}} = \frac{1}{r} \frac{\partial U}{\partial r} \boldsymbol{r}$$

Let's apply the transformation of coordinates

$$\boldsymbol{r} \to e^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}} \mathfrak{r}, \qquad \boldsymbol{\varphi} = \boldsymbol{\Omega} t, \qquad (T^a)_{bc} = i\varepsilon_{bac}, \qquad \boldsymbol{r}^2 = \mathfrak{r}^2$$

As You may know, $e^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}}$ is a rotation matrix, and $\boldsymbol{T}$ are generators of the rotation group, so

$$\left(e^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}}\right)_{ab} = n_a n_b + \cos\varphi\, (\delta_{ab} - n_a n_b) + \varepsilon_{akb} n^k \sin\varphi$$

Take derivatives d/dt, d^2/dt^2 using commutativity of $\boldsymbol{\Omega}\cdot\boldsymbol{T}$ with $e^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}}$:

$$\frac{de^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}}}{dt} = e^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}}\, i\boldsymbol{\Omega}\cdot\boldsymbol{T}$$

Problem 1 (solution)

$$\dot{\boldsymbol{r}} \to e^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}}\left[i\left(\dot{\boldsymbol{\varphi}}\cdot\boldsymbol{T}\right)\mathfrak{r}+\dot{\mathfrak{r}}\right]$$

$$\ddot{\boldsymbol{r}} \to e^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}}\left[i\left(\ddot{\boldsymbol{\varphi}}\cdot\boldsymbol{T}\right)\mathfrak{r}+\ddot{\mathfrak{r}}+2i\left(\dot{\boldsymbol{\varphi}}\cdot\boldsymbol{T}\right)\dot{\mathfrak{r}}-\left(\dot{\boldsymbol{\varphi}}\cdot\boldsymbol{T}\right)\left(\dot{\boldsymbol{\varphi}}\cdot\boldsymbol{T}\right)\mathfrak{r}\right]=$$

$$=e^{i\boldsymbol{\varphi}\cdot\boldsymbol{T}}\left[\ddot{\mathfrak{r}}+2i\left(\boldsymbol{\Omega}\cdot\boldsymbol{T}\right)\dot{\mathfrak{r}}-\left(\boldsymbol{\Omega}\cdot\boldsymbol{T}\right)\left(\boldsymbol{\Omega}\cdot\boldsymbol{T}\right)\mathfrak{r}\right]$$

Now write out everything in components:

$$i\left(\boldsymbol{\Omega}\cdot\boldsymbol{T}\right)\dot{\mathfrak{r}}=-\varepsilon_{anb}\Omega_n\dot{\mathfrak{r}}_b=-\left[\boldsymbol{\Omega}\times\dot{\mathfrak{r}}\right]_a$$

$$-\left(\boldsymbol{\Omega}\cdot\boldsymbol{T}\right)\left(\boldsymbol{\Omega}\cdot\boldsymbol{T}\right)\mathfrak{r}=\left[\boldsymbol{\Omega}\times\left[\boldsymbol{\Omega}\times\mathfrak{r}\right]\right]_a$$

after substitution of these results all Ω-dependent terms cancel and we end up with

$$m\,\ddot{\mathfrak{r}}=-\boldsymbol{\nabla}U,\qquad \boldsymbol{\nabla}U=\frac{1}{\mathfrak{r}}\frac{\partial U}{\partial\mathfrak{r}}\mathfrak{r}$$

i.e. the equation for motion in inertial frame (as You may guess, the variables $\mathfrak{r}$ are the coordinates in the inertial frame)

⇒In non-inertial frame the trajectory will rotate with velocity $\boldsymbol{\Omega}$

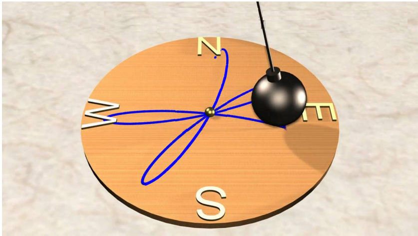
Problem 1a

Assume that You make a mechanical experiment and analyze motion on the 2D horizontal surface in the field of the axially symmetric potential field $U(r,z)$. Analyze how the rotation of the Earth will the trajectory of particles

Note: *Many 3D problems in presence of constraints reduce to effective 2D motion in axially symmetric potentials:*

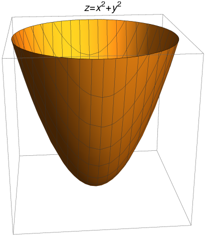
Example: Foucault's pendulum, constraint $z^2 = \ell^2 - x^2 - y^2$

$$U = mgz \approx \frac{mg}{2\ell}\left(x^2 + y^2\right)$$



Example: Motion over any axially symmetric surface $z = f(r)$ in gravit. field

$$U = mgz = mgf(r)$$



Problem 1a (solution)

Assume $\boldsymbol{\Omega} = \Omega\, \boldsymbol{n} = \mathrm{const}$ is the vector of angular velocity
Assume $\hat{\boldsymbol{z}}$ is normal to the surface, so radius-vector (position) $\boldsymbol{r} = (x, y, 0)$
Equations of motion in non-inertial frame ():

$$m\,\ddot{\boldsymbol{r}} = -2m\boldsymbol{\Omega} \times \boldsymbol{v} - m\,\boldsymbol{\Omega} \times [\boldsymbol{\Omega} \times \boldsymbol{r}] - \boldsymbol{\nabla} U, \qquad \boldsymbol{\nabla} U = \frac{\partial U}{\partial r}\hat{\boldsymbol{r}} = \frac{1}{r}\frac{\partial U}{\partial r}\boldsymbol{r},$$

Let's apply the transformation of coordinates

$$\boldsymbol{r} \to e^{i\varphi \hat{\boldsymbol{z}}\cdot \boldsymbol{T}}\mathfrak{r}, \qquad \varphi = \Omega_z t, \qquad (T^a)_{bc} = i\varepsilon_{bac}, \qquad \boldsymbol{r}^2 = \mathfrak{r}^2$$

As You may know, $e^{i\varphi \hat{\boldsymbol{z}}\cdot \boldsymbol{T}}$ is a rotation matrix around axis z
Take derivatives d/dt, d^2/dt^2 :

$$\frac{de^{i\varphi \hat{\boldsymbol{z}}\cdot \boldsymbol{T}}}{dt} = e^{i\varphi \hat{\boldsymbol{z}}\cdot \boldsymbol{T}} i\Omega_z \hat{\boldsymbol{z}} \cdot \boldsymbol{T}$$

Problem 1a (solution)

$$\dot{\boldsymbol{r}} \to e^{i\varphi \hat{\boldsymbol{z}}\cdot\boldsymbol{T}}\left[i\left(\dot{\varphi}\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\mathfrak{r} + \dot{\mathfrak{r}}\right]$$

$$\ddot{\boldsymbol{r}} \to e^{i\varphi \hat{\boldsymbol{z}}\cdot\boldsymbol{T}}\left[i\left(\ddot{\varphi}\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\mathfrak{r} + \ddot{\mathfrak{r}} + 2i\left(\dot{\varphi}\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\dot{\mathfrak{r}} - \dot{\varphi}^2\left(\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\left(\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\mathfrak{r}\right] =$$

$$= e^{i\varphi \hat{\boldsymbol{z}}\cdot\boldsymbol{T}}\left[\ddot{\mathfrak{r}} + 2i\Omega\left(\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\dot{\mathfrak{r}} - \Omega^2\left(\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\left(\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\mathfrak{r}\right]$$

Now write out everything in components:

$$i\left(\Omega\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\dot{\mathfrak{r}} = -\varepsilon_{azb}\Omega\dot{\mathfrak{r}}_b = -\Omega\left[\hat{\boldsymbol{z}}\times\dot{\mathfrak{r}}\right]_a$$

$$-\left(\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\left(\hat{\boldsymbol{z}}\cdot\boldsymbol{T}\right)\mathfrak{r} = \left[\hat{\boldsymbol{z}}\times\left[\hat{\boldsymbol{z}}\times\mathfrak{r}\right]\right]_a$$

after substitution of these results all Ω-dependent terms cancel and we end up with

$$m\ddot{\mathfrak{r}} = -\boldsymbol{\nabla}U, \qquad \boldsymbol{\nabla}U = \frac{1}{\mathfrak{r}}\frac{\partial U}{\partial\mathfrak{r}}\mathfrak{r}$$

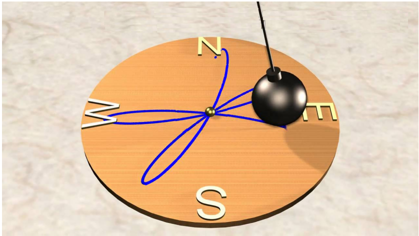
i.e. the equation for motion in inertial frame
$\Rightarrow$The trajectory will rotate around axis $\hat{\boldsymbol{z}}$ with angular velocity $\Omega_z \leq \Omega$

Corollary

Many 3D problems in presence of constraints reduce to effective 2D motion in axially symmetric potentials

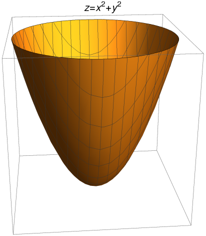
Example: Foucault's pendulum, constraint $z^2 = \ell^2 - x^2 - y^2$

$$U = mgz \approx \frac{mg}{2\ell}\left(x^2 + y^2\right)$$



Example: Motion over any axially symmetric surface $z = f(r)$

$$U = mgz = mgf(r)$$



We may find the trajectory in inertial frame, and after that rotate it with angular velocity $\Omega_z \leq \Omega$ to get in non-inertial reference frame

Problem 1b

Now we will show this explicitly, solving the equations of motion for small oscillations of pendulum near minimum

- We'll assume that the lagrangian in inertial frame is given by

$$L \approx \frac{m}{2}\left[\dot{x}^2 + \dot{y}^2\right] - \frac{k}{2}\left[x^2 + y^2\right]$$

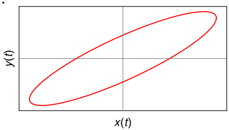
- Later we'll introduce fictitious forces and solve the equations of motion explictly.

Problem 1b

► Lagrangian in Cartesian coordinates in inertial frame:

$$L \approx \frac{m}{2}\left[\dot{x}^2 + \dot{y}^2\right] - \frac{k}{2}\left[x^2 + y^2\right]$$

so the motion is just two independent oscillations in x and y directions with frequency $\omega = \sqrt{k/m} = \sqrt{g/\ell}$
− The trajectory is an ellipse:



$$\begin{aligned} x &= A_x \cos\left(\omega t + \phi_x\right), \\ y &= A_y \cos\left(\omega t + \phi_y\right). \end{aligned}$$

► Effects of the Earth rotation are small, can disregard $\mathcal{O}\left(\Omega^2\right)$-terms (centrifug. forces), take into account only Coriolis:

$$m\,\ddot{\boldsymbol{r}} = -k\,\boldsymbol{r} + 2m\,\boldsymbol{\Omega} \times \dot{\boldsymbol{r}}$$

▷ Motion in vertical direction $\dot{z} \sim A_x^2,\ A_y^2$, negligible compared to $\dot{x} \sim A_x,\ \dot{y} \sim A_y$.

$$\begin{aligned} m\,\ddot{x} &= -k\,x - 2m\,\Omega_z \dot{y}, \\ m\,\ddot{y} &= -k\,y + 2m\,\Omega_z \dot{x}, \end{aligned}$$

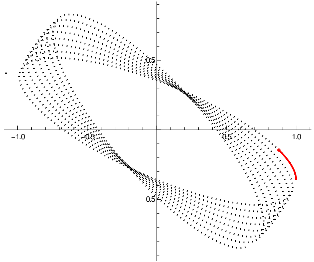
Problem 1b

Introduce variable $\xi = x + iy$. In terms of this variable, the equation has a form

$$\ddot{\xi} + 2i\Omega\dot{\xi} + \omega^2\xi = 0$$

We can eliminate the $\sim \dot{\xi}$ term defining new variable ζ:

$$\xi = \zeta e^{-i\Omega t}, \quad \dot{\xi} = \left(\dot{\zeta} - i\Omega\zeta\right) e^{-i\Omega t}, \quad \ddot{\xi} = \left(\ddot{\zeta} - 2i\Omega\dot{\zeta} - \Omega^2\zeta\right) e^{-i\Omega t}$$



$$\ddot{\zeta} + \left(\omega^2 - \Omega^2\right)\zeta = 0$$

For ζ we recovered equation for free oscillations,

$$\begin{aligned} \mathrm{Re}\zeta &= \mathcal{A}_x \cos\left(\tilde{\omega}t + \phi_x\right), \\ \mathrm{Im}\zeta &= \mathcal{A}_y \cos\left(\tilde{\omega}t + \phi_y\right) \\ \tilde{\omega} &= \sqrt{\omega^2 - \Omega^2}. \end{aligned}$$

Result for ξ (take real and imaginary parts):

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \mathrm{Re}(\xi) \\ \mathrm{Im}(\xi) \end{pmatrix} = \begin{pmatrix} \cos\Omega t & \sin\Omega t \\ -\sin\Omega t & \cos\Omega t \end{pmatrix} \begin{pmatrix} \mathrm{Re}(\zeta) \\ \mathrm{Im}(\zeta) \end{pmatrix}$$

-effectively we have rotation of the trajectory $\zeta(t)$ with angular velocity Ω

Problem 2

Deviations from vertical fall

- The body falls freely from the height h with zero initial velocity. Evaluate the deflection from the vertical due to the Earth rotation. Assume that the angular velocity Ω is small.

Problem 2 (solution)

The centrifugal force is higher order in Ω^2 and can be neglected compared to Coriolis. Equation of motion has a form

$$\ddot{\boldsymbol{r}} = \boldsymbol{g} + 2\,\boldsymbol{v} \times \boldsymbol{\Omega} = \boldsymbol{g} + 2\,\dot{\boldsymbol{r}} \times \boldsymbol{\Omega} \qquad (1)$$

We consider the second term as a perturbation and develop a perturbation theory:
−We assume that the solution has a form

$$\boldsymbol{r} = \boldsymbol{r}^{(0)} + \underbrace{\boldsymbol{r}^{(1)}}_{\sim|\Omega|} + \underbrace{\boldsymbol{r}^{(2)}}_{\sim|\Omega|^2} + \ldots,$$

where $\boldsymbol{r}^{(0)}$ is the solution of (1) with $\Omega = 0$ (free fall), $\boldsymbol{r}^{(1)}$ is a small linear correction, etc.
−We request that the equation (1) is satisfied order-by-order, separately for terms $\sim \Omega^0$, $\sim \Omega^1, \ldots$

$$\ddot{\boldsymbol{r}}^{(0)} = \boldsymbol{g} \qquad (1a)$$

$$\ddot{\boldsymbol{r}}^{(1)} = 2\,\dot{\boldsymbol{r}}^{(0)} \times \boldsymbol{\Omega} \qquad (1b)$$

....

Problem 2 (solution)

$$
\begin{aligned}
\ddot{\boldsymbol{r}}^{(0)} &= \boldsymbol{g} && (1a) \\
\ddot{\boldsymbol{r}}^{(1)} &= 2\,\dot{\boldsymbol{r}}^{(0)} \times \boldsymbol{\Omega} && (1b) \\
&\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots
\end{aligned}
$$

The integration of (1a, 1b, ...) is straightforward: in right-hand side have a function known from previous iteration, so just need to take 2 integrals:

Corresponding solutions:

$$
\boldsymbol{r}^{(0)} = \boldsymbol{h} + \frac{\boldsymbol{g}\,t^2}{2}, \qquad \ddot{\boldsymbol{r}}^{(1)} = 2\,\dot{\boldsymbol{r}}^{(0)} \times \boldsymbol{\Omega} \Rightarrow \boldsymbol{r}^{(1)} = \boldsymbol{g} \times \boldsymbol{\Omega}\,\frac{t^3}{3}, \ldots
$$

Substituting the time of flight $t_f = \sqrt{2h/g}$, we may obtain the full deviation

$$
|\delta r| \equiv \left|\boldsymbol{r}^{(1)}\left(t_f\right)\right| = \frac{1}{3}\left(\frac{2h}{g}\right)^{3/2} g\,\Omega\,|\cos\theta|
$$

where θ is latitude ($\theta = 0$ at the Equator).

Rigid body motion

—We'll start with introductory questions ...

Control question 1

What is a rigid body ?
How many degrees of freedom it has?

Control question 2

How can/should we choose the degrees of freedom which describe rotations?

Control question 3

How does the rotation affect the motion of the body?
– Can we assume that rotation and motion of the center-of-mass are independent?

Motion of rigid body

Rigid body

Rigid body=system of masses which moves "as a whole", distance between any pair of points=const

- Six degrees of freedom:

– Translations: $\boldsymbol{r} \to \boldsymbol{r} + \boldsymbol{a}$

– (infinitesimal) rotations: $\boldsymbol{r} \to \boldsymbol{r} + \boldsymbol{\phi} \times \boldsymbol{r}$

– (Finite) rotations: $r_a \to R_{ab}(\phi) r_b$

–Rotations should NOT change scalar products $(A \cdot B)$ and norms of vectors $(A \cdot A)$ $\Rightarrow$Matrices $R \in \mathrm{SO}(3)$:

$$R^T R = 1, \qquad \left(R^T\right)_{ij} = R_{ji}, \qquad \det R = +1$$

–For small rotations $R_{ab} \approx \delta_{ab} + \sum_c \epsilon_{acb}\phi^c = \delta_{ab} + i\sum_c \left(T_c\right)_{ab}\phi^c, \quad T_c = -i\epsilon_{acb}$

- In this course we'll see that sometimes consideration is simpler when we work in the non-inertial body's reference frame

  –In that frame body's shape/distribution of mass is constant

Euler's theorem

► Any motion of rigid body with one point fixed is rotation around some axis
Equivalent formulation:
−The rotation matrix always has at least one eigenvalue=1,

$$\hat{R}\,\boldsymbol{A} = \lambda\,\boldsymbol{A}, \qquad R_{ij}A_j = \lambda\,A_i$$

Proof:

$$R^{\mathrm{T}}R = \hat{1} \Rightarrow \det R = \det R^{\mathrm{T}} = +1$$

$$R^{\mathrm{T}}\left(R - \hat{1}\right) = \hat{1} - R^{\mathrm{T}} = (1 - R)^{\mathrm{T}}$$

Now take $\det(...)$ of both parts, take into account that for any M, $\det M = \det M^{\mathrm{T}}$

$$\underbrace{\det R}_{1}\,\det\left(R - \hat{1}\right) = \det\left(\hat{1} - R\right)$$

In 3 dimensions $\det(-M) = -\det(M)$, so get $\det\left(R - \hat{1}\right) = 0$
⇒At least one eigenvalue of $(R - \hat{1})$ should vanish.
−Since R and $(R - \hat{1})$ have the same eigenvectors, at least eigenvalue λ_1 of R should be $\lambda_1 = 1$. Q.E.D.

Rotation matrix

– If λ_1, λ_2, λ_3 are eigenvalues of R, in view of $\det R = \lambda_1\lambda_2\lambda_3 = +1$ and $\lambda_1 = 1$ conclude that $\lambda_2\lambda_3 = +1$

– In general λ_2, λ_3 and eigenvectors $\boldsymbol{e}_2, \boldsymbol{e}_3$ of R can be complex.

$$\hat{R}\boldsymbol{e}_a = \lambda_a \boldsymbol{e}_a \quad \Rightarrow \quad \left|\hat{R}\boldsymbol{e}_a\right|^2 = |\lambda_a \boldsymbol{e}_a|^2 = |\lambda_a|^2$$

but

$$\left|\hat{R}\boldsymbol{e}_a\right|^2 = \left(\hat{R}\boldsymbol{e}_a\right)^* \left(\hat{R}\boldsymbol{e}_a\right) = \boldsymbol{e}_a^* \hat{R}^{\mathrm{T}} \hat{R}\boldsymbol{e}_a = \boldsymbol{e}_a^* \boldsymbol{e}_a = 1$$

$$\Rightarrow |\lambda_a|^2 = 1, \qquad \lambda_2 = e^{i\Phi},\ \lambda_3 = e^{-i\Phi}, \quad \Phi \in \mathbb{R}$$

$$\mathrm{Tr}\hat{R} = 1 + e^{i\Phi} + e^{-i\Phi} = 1 + 2\cos\Phi$$

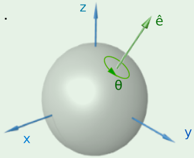
Parametrization of rotations

▶For $N = 3$ (our real space) we need just 3 parameters - "rotation angles"
−We'll use vectorial notation (e.g. $\vec{\alpha}$) for any such set of the 3 numbers

▶There is no "a priori" parametrization for R, $\underline{any}$ set of 3 parameters could be used (like choice of curvilinear coordinates in 3D space).
−In physics we use parametrizations which:
1) have clear physical meaning, and
2) allow to get relatively simple equations of motion

Common parametrizations of rotation matrix

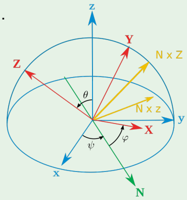
Axis-Angle convention



- Axis $\vec{e}\,(\phi,\,\psi)$
- Angle θ

$$\boldsymbol{\theta} = \theta\, \boldsymbol{e}$$

Euler angles



$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$$

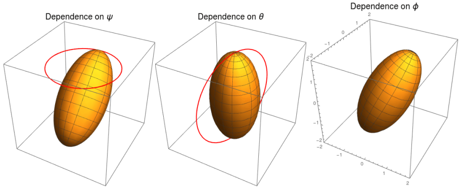
$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Superposition of "elementary" rotations

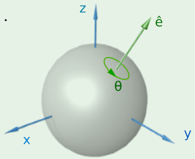
$$R\,(\psi, \theta, \varphi) = R_Z\,(\varphi)\, R_X\,(\theta)\, R_Z\,(\psi)$$

we'll use *ZXZ* convention for definiteness



Control problem

Axis-Angle convention



Assume that You know a rotation matrix $\hat{R}$ in some convention, and would like to rewrite it in Axis-Angle convention, in terms of rotation axis $\boldsymbol{e}$ and angle θ (so $\boldsymbol{\theta} = \theta \boldsymbol{e}$). How can we find the rotation axis $\boldsymbol{e}$ and rotation angle θ?

Help: *It is purely mathematical problem:*

1) What happens with $\mathbf{e}$ *when You apply rotation* $\hat{R}$ *to it ?*

– How this can be used to find $\mathbf{e}$*?*

2) Assume we choose axis $\hat{\mathbf{z}}$ *in direction of* $\mathbf{e}$ *and evaluate trace* $\mathrm{Tr}\left(\hat{R}\right)$.

– What will be the result?

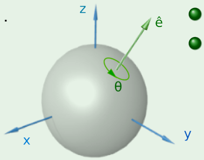
– Assume You now applied global transformation

$$\hat{R} \to S\hat{R}S^{-1}.$$

(change of basis). Will the trace $\mathrm{Tr}\left(\hat{R}\right)$ *change after such transformation?*

Control problem (reply)

Axis-Angle convention



- Axis $\boldsymbol{e}\,(\phi,\,\psi)$
- Angle θ

$$\boldsymbol{\theta} = \theta\,\boldsymbol{e}$$

Matrix $\hat{R}$→Axis-angle:

- Rotation axis $\boldsymbol{e}$ is eigenvector of rotation matrix with unit eigenvalue:

$$\hat{R}\,\boldsymbol{e} = \boldsymbol{e}$$

- For any rotation matrix angle θ is given by

$$\mathrm{Tr}\left(\hat{R}\right) = 1 + 2\cos\theta$$

Control problem 2

How can we construct explicitly rotation matrix $\hat{R}$ if we know $\boldsymbol{e}, \theta$ in Axis-angle convention?

Control problem 2 (solution)

Any finite rotation around axis $\boldsymbol{n}$ by angle θ might be represented as a superposition of elementary rotations,

$$\hat{R}(\theta) = \left[\hat{R}\left(\frac{\theta}{N}\right)\right]^N, \qquad N \gg 1 \qquad (1)$$

For small angles $\phi = \theta/N \ll 1$ we have transformation of vector components

$$\boldsymbol{A} \qquad \to \boldsymbol{A}' = \boldsymbol{A} + \boldsymbol{\phi} \times \boldsymbol{A}$$

or in matrix form

$$A_a \to A'_a = \hat{R}_{ab}(\phi)\, A_b \approx A_a + \epsilon_{abc}\phi^b A^c$$

$$\Rightarrow R_{ab} \approx \delta_{ab} + \sum_c \phi^c \varepsilon_{acb} = \delta_{ab} + i \sum_c \phi^c (T_c)_{ab}, \quad (T_c)_{ab} = -i\varepsilon_{acb} \qquad (2)$$

From (1,2) conclude that

$$R(\theta) = \lim_{N\to\infty}\left(1 + \frac{1}{N} i \sum_c T_c \theta^c\right)^N = e^{i\sum_c T_c\theta^c} \qquad (3)$$

In theory of Lie groups matrices T_c used for expansion are called generators, and expression (3) is a fundamental result known as Lie's theorem

Control problem 2 (solution)

$$R(\theta) = e^{i\sum_c T_c \theta^c} = \sum_{k=0}^{\infty} \frac{1}{k!}\left(i \sum_c T_c \theta^c\right)^k \qquad (3)$$

To evaluate (3), we may use Taylor expansion:
Using property of Levi-Civita symbols

$$\varepsilon_{kab}\varepsilon_{kcd} = \delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}$$

we may obtain that

$$(n \cdot T)^2{}_{ab} = \delta_{ab} - n_a n_b,$$

Similarly, for even powers of $n \cdot T$:

$$\begin{aligned}
(n \cdot T)^4 \quad {}_{ab} &= (n \cdot T)^2 (n \cdot T)^2 = (n \cdot T)^2 = \delta_{ab} - n_a n_b, \\
(n \cdot T)^6 \quad {}_{ab} &= (n \cdot T)^2 (n \cdot T)^4 = (n \cdot T)^2 = \delta_{ab} - n_a n_b, \\
(n \cdot T)^{2\ell} \quad {}_{ab} &= \delta_{ab} - n_a n_b
\end{aligned}$$

For odd powers of $n \cdot T$ we may also get a similar compact result,

$$(n \cdot T)_{ab} = (n \cdot T)_{ac} (\delta_{cb} - n_c n_b),$$

$$\left[(n \cdot T)^{2\ell+1}\right]_{ab} = \left[(n \cdot T)(n \cdot T)^{2\ell+1}\right]_{ab} = (n \cdot T)_{ac} (\delta_{cb} - n_c n_b),$$

Control problem 2 (solution)

$$
\begin{aligned}
R_{ab} &= \sum_k \frac{i^k\theta^k}{k!}\left(n\cdot T\right)^k = \sum_{k=\text{odd}} \frac{i^k\theta^k}{k!}\left(n\cdot T\right)^k + \sum_{k=\text{even}} \frac{i^k\theta^k}{k!}\left(n\cdot T\right)^k \\
&= \underbrace{\sum_{\ell=0}^{\infty} \frac{i^{2\ell+1}\theta^{2\ell+1}}{(2\ell+1)!}}_{\sin(\theta)} \underbrace{\left(n\cdot T\right)^{2\ell+1}}_{=(n\cdot T)} + \sum_{\ell=0}^{\infty} \frac{i^{2\ell}\theta^{2\ell}}{(2\ell)!} \underbrace{\left(n\cdot T\right)^{2\ell}}_{=(n\cdot T)^2}
\end{aligned}
$$

The constant in front of $\sim (n\cdot T)^2$ "looks like" $\cos\theta$, yet we should be careful for $\ell = 0$:
– can see that $(n\cdot T)^{2\ell} = 1$, whereas all other terms are $\sim (n\cdot T)^2 \neq 1$. So let's rewrite it as

$$
(n\cdot T)^0 \equiv 1 = 1 - (n\cdot T)^2 + (n\cdot T)^2
$$

which allows to make proper resummations and rewrite the result as:

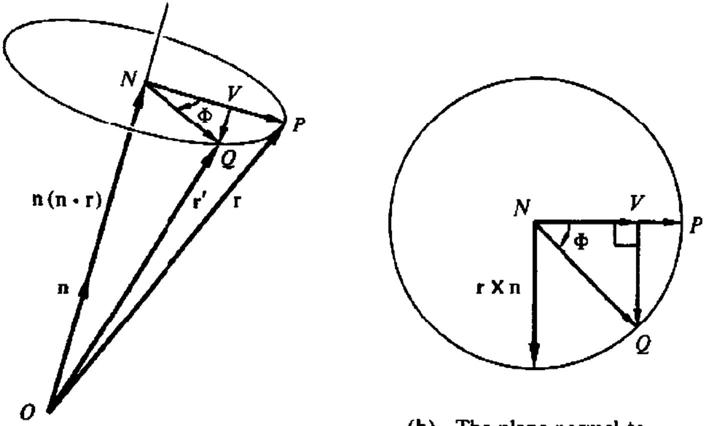
$$
\begin{aligned}
R_{ab} &= n_a n_b + \cos\theta\,\left(\delta_{ab} - n_a n_b\right) + i\left(n\cdot T\right)_{ab}\sin\theta \\
&= n_a n_b + \cos\theta\,\left(\delta_{ab} - n_a n_b\right) + \varepsilon_{akb} n^k \sin\theta
\end{aligned}
$$

Control problem 2 (solution)

Geometrical interpretation (from Goldstein)

$$R_{ab} = n_a n_b + \cos\Phi\,(\delta_{ab} - n_a n_b) + \varepsilon_{akb} n^k \sin\Phi$$

$$\hat{R}(\Phi, \boldsymbol{n})\,\boldsymbol{r} = \boldsymbol{n}\,(n \cdot r) + \cos\Phi\,(\boldsymbol{r} - \boldsymbol{n}\,(n \cdot r)) + \boldsymbol{r} \times \boldsymbol{n} \sin\Phi$$



(a) Overall view

(b) The plane normal to the axis of rotation

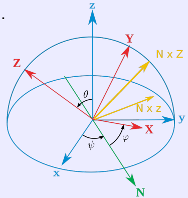
Control problem 3

Euler angles→Matrix R

–If we know angles ψ, θ, φ, by definition can construct the rotation matrix R as product of elementary rotations

$$R(\psi, \theta, \varphi) = R_Z(\varphi) R_X(\theta) R_Z(\psi)$$

Matrix R →Euler angles (Inverse problem)



Assume that You know a rotation matrix $\hat{R}$ in some convention, and would like to rewrite it in terms of Euler angles (ZXZ) convention. How can we find these angles ψ, θ, φ ?

Control problem 3

Matrix→Euler

- If we know $\hat{R}$, can get Euler angles using simple algebra.

E.g. for ZXZ convension we have

$$R = Z_1 X_2 Z_3 = \begin{bmatrix} c_1 c_3 - c_2 s_1 s_3 & -c_1 s_3 - c_2 c_3 s_1 & s_1 s_2 \\ c_3 s_1 + c_1 c_2 s_3 & c_1 c_2 c_3 - s_1 s_3 & -c_1 s_2 \\ s_2 s_3 & c_3 s_2 & c_2 \end{bmatrix}$$

where $c_i = \cos\theta_i$, $s_i = \sin\theta_i$, $\theta_{1,2,3} = (\phi, \theta, \psi)$

$$\Rightarrow \psi = \arctan\left(R_{31}/R_{32}\right)$$
$$\Rightarrow \theta = \arccos\left(R_{33}\right)$$
$$\Rightarrow \phi = -\arctan\left(R_{13}/R_{23}\right)$$

Important remark

In the previous discussions we assumed that rotation vector $\boldsymbol{\theta}$ (or Euler angles ψ, θ, ϕ) are constants.

In dynamical problems which involve rotation of the body, these variables ($\boldsymbol{\theta}$ or ψ, θ, ϕ) become functions of time. Even direction of $\boldsymbol{\theta}$ is NOT constant, depends on time

—Coordinates $r_b^{(\mathrm{body})}$ of any fixed element in the body's rest frame are related to its coordinates in the lab frame as

$$r_a^{(\mathrm{lab})}(t) = R_{ab}\left(\psi(t),\, \theta(t),\, \varphi(t)\right) r_b^{(\mathrm{body})}$$

*The density $\rho = \rho\left(\boldsymbol{r}^{(\mathrm{body})}\right)$ does not depend on time in the body's rest frame

** ... but depends on time when we write it in terms of $\boldsymbol{r}^{(\mathrm{lab})}$

Important remark

—Similarly, if body moves as a whole (without rotation), its coordinates in body's rest frame and lab frame are related as

$$r_a^{(\mathrm{lab})}(t) = r_a^{(\mathrm{body})} + X_a(t), \quad a = 1,2,3 \ (a = x, y, z)$$

where $X_a(t)$ are coordinates of in the body's center-of-mass ...

*The density $\rho = \rho\left(\boldsymbol{r}^{(\mathrm{body})}\right)$ does not depend on time in the body's rest frame

** ... but is NOT const at any fixed point in lab frame $\boldsymbol{r}^{(\mathrm{lab})}$

—As we discussed earlier, $\psi(t)$, $\theta(t)$, $\phi(t)$, $X_a(t)$ should be treated as degrees of freedom (rotations & translations)

*Time dependence should be fixed from (Euler-Lagrange) equations of motion & initial conditions

- We learned how to parametrize the rotation matrix and use it to transform different vectors.
- Now we can return back to discussion of physics in inertial vs body's rest frame.

Rotation around axis

Body rotates only around one axis, $\boldsymbol{\theta} = \theta(t)\,\boldsymbol{n}$, where $\boldsymbol{n} = \mathrm{const}$ (i.e. direction of rotation axis is conserved).

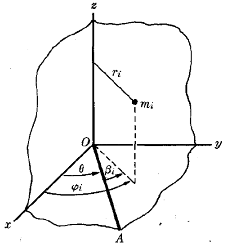
$-$Angular (rotational) dynamics is described by:

$$\frac{dM}{dt} = K^{(\mathrm{ext})}, \quad M = I\,\dot{\theta}$$

where K is torque,

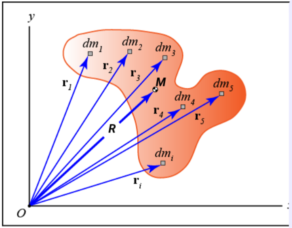
$I = \mathrm{const}$ is a moment of inertia,

$$I = \sum m_i r_{i\perp}^2 = \int dV\,\rho\,r_\perp^2$$



In general for rotation around one axis, I does not depend on time. However, for bodies of irregular shape I depends on the orientation of the rotation axis (which determines $\boldsymbol{r}_\perp$)

Kinetic energy of rigid body



Assume that the rigid body rotates with constant angular velocity $\boldsymbol{\Omega}$ around axis $\boldsymbol{n}$ passing through its center of mass (direction of vector $\boldsymbol{\Omega}$ coincides with rotation axis, and $|\boldsymbol{\Omega}|$ is the absolute value of angular velocity). Also, the center of mass of the body moves with velocity $\boldsymbol{V}$ in the lab frame. Find the kinetic energy of the rigid body. Demonstrate that kinetic energy may be separated into 2 pieces,

$$T = T_{\rm CM} + T_{\rm rotation}$$

- Assume that at the moment $t = 0$ the position of element dm_i is given by vector $\boldsymbol{r}$
  - At the moment $t + dt$, the position is given by vector $\boldsymbol{r}'$ slightly displaced w.r.t. $\boldsymbol{r}$:
    - For infinitesimal rotation angles $\boldsymbol{\theta} \ll 1$ we may write the expressions for displacements of each element dm_i and corresponding velocities as

$$\delta\boldsymbol{r} = \boldsymbol{r}' - \boldsymbol{r} = \boldsymbol{\theta}(t) \times (\boldsymbol{r} - \boldsymbol{R}_{\rm CM}) + \delta\boldsymbol{R}_{\rm CM}(t),$$
$$\boldsymbol{v} \equiv \delta\dot{\boldsymbol{r}} = \dot{\boldsymbol{\theta}} \times (\boldsymbol{r} - \boldsymbol{R}_{\rm CM}) + \boldsymbol{V} = \boldsymbol{\Omega} \times (\boldsymbol{r} - \boldsymbol{R}_{\rm CM}) + \boldsymbol{V}$$

Tensor of inertia

$$\boldsymbol{v} = \boldsymbol{\Omega} \times (\boldsymbol{r} - \boldsymbol{R}_{\mathrm{CM}}) + \boldsymbol{V}, \qquad \boldsymbol{R} = \frac{\int \rho(\boldsymbol{r})\, \boldsymbol{r}\, dV}{M}, \quad M \equiv \int \rho(\boldsymbol{r})\; dV$$

Kinetic energy:

$$T = \frac{1}{2} \int d^3 r\, \rho(\boldsymbol{r}) \boldsymbol{v}^2(\boldsymbol{r}, t) =$$

$$\frac{1}{2} M \boldsymbol{V}^2 + \frac{1}{2} \int d^3 r\, \rho(r) \left(\boldsymbol{\Omega} \times (\boldsymbol{r} - \boldsymbol{R}_{\mathrm{CM}})\right)^2 + \boldsymbol{V} \cdot \left(\boldsymbol{\Omega} \times \underbrace{\int d^3 r\, \rho(r)\, (\boldsymbol{r} - \boldsymbol{R}_{\mathrm{CM}})}_{0} \right)$$

$$= \underbrace{\frac{1}{2} M \boldsymbol{V}^2}_{\text{CM motion}} + \frac{1}{2} \Omega_j \Omega_k \underbrace{\int dV\, \rho(r) \left(\delta_{jk} \mathfrak{r}^2 - \mathfrak{r}_j \mathfrak{r}_k\right)}_{I_{jk}}, \qquad \boldsymbol{\mathfrak{r}} \equiv \boldsymbol{r} - \boldsymbol{R}_{\mathrm{CM}}$$

I_{jk} is called the tensor of inertia

—We'll derive the red-colored expression in the next slide

—Kinetic energy indeed separates into 2 pieces, CM motion and rotation.

$$T = T_{\mathrm{CM}} + T_{\mathrm{rotation}}$$

Technical remark on vector algebra

Our goal

prove that

$$[\boldsymbol{A} \times \boldsymbol{B}] \cdot [\boldsymbol{C} \times \boldsymbol{D}] = (A \cdot C)(B \cdot D) - (A \cdot D)(B \cdot C) \qquad (1)$$

and

$$[\boldsymbol{\Omega} \times \boldsymbol{r}]^2 = \boldsymbol{\Omega}^2 \boldsymbol{r}^2 - (\boldsymbol{\Omega} \cdot \boldsymbol{r})^2$$

- Method 1 (without use of Levi-Civita symbol):

$$\boldsymbol{R}_1 = \boldsymbol{A} \times \boldsymbol{B} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}$$

$$\boldsymbol{R}_2 = \boldsymbol{C} \times \boldsymbol{D} = \begin{vmatrix} \hat{\boldsymbol{x}} & \hat{\boldsymbol{y}} & \hat{\boldsymbol{z}} \\ C_x & C_y & C_z \\ D_x & D_y & D_z \end{vmatrix}$$

$\Rightarrow$So we need to write out explicitly vector products ($\times$), after that evaluate dot-product $R_1 \cdot R_2$ and demonstrate that result coincides with right-hand side of (1) (difficult but doable with Mathematica)

- Method 2 (without use of Levi-Civita symbol):

$$[\boldsymbol{A} \times \boldsymbol{B}] \cdot \boldsymbol{Z} = \boldsymbol{A} \cdot [\boldsymbol{B} \times \boldsymbol{Z}]$$

$$\boldsymbol{Z} \to \boldsymbol{C} \times \boldsymbol{D}$$

$$\boldsymbol{B} \times [\boldsymbol{C} \times \boldsymbol{D}] = \boldsymbol{C}\,(B \cdot D) - \boldsymbol{D}\,(B \cdot C)$$

$$\Downarrow$$

$$[\boldsymbol{A} \times \boldsymbol{B}] \cdot [\boldsymbol{C} \times \boldsymbol{D}] = (A \cdot C)(B \cdot D) - (A \cdot D)(B \cdot C)$$

Technical remark on vector algebra

$$[\boldsymbol{A} \times \boldsymbol{B}] \cdot [\boldsymbol{C} \times \boldsymbol{D}] = \qquad (A \cdot C)(B \cdot D) - (A \cdot D)(B \cdot C) \quad (1)$$

If we substitute $\boldsymbol{A}$, $\boldsymbol{C} \to \boldsymbol{\Omega}$ and $\boldsymbol{B}, \boldsymbol{D} \to \boldsymbol{r}$, we'll get

$$[\boldsymbol{\Omega} \times \boldsymbol{r}]^2 = \boldsymbol{\Omega}^2 \boldsymbol{r}^2 - (\boldsymbol{\Omega} \cdot \boldsymbol{r})^2$$

In components (Cartesian coordinates):

$$\Omega^2 = \sum_{ij} \Omega_i \Omega_j \delta_{ij}$$

$$(\boldsymbol{\Omega} \cdot \boldsymbol{r}) = \sum_i \Omega_i r_i,$$

$$(\boldsymbol{\Omega} \cdot \boldsymbol{r})^2 = (\boldsymbol{\Omega} \cdot \boldsymbol{r})(\boldsymbol{\Omega} \cdot \boldsymbol{r}) = \sum_i \Omega_i r_i \sum_j \Omega_j r_j = \sum_{ij} \Omega_i \Omega_j r_i r_j$$

$$\Rightarrow [\boldsymbol{\Omega} \times \boldsymbol{r}]^2 = \sum_{ij} \Omega_i \Omega_j \left(\delta_{ij} r^2 - r_i r_j\right)$$

Properties of tensor of inertia

$$I_{jk} = \int d^3r\, \rho(\boldsymbol{r}, t) \left(\delta_{jk}\mathfrak{r}^2 - \mathfrak{r}_j \mathfrak{r}_k\right)$$

Some basic properties of the tensor of inertia I_{jk}:
−Symmetric 3×3 matrix
−Positively defined
−Depends on orientation (via $\rho(\boldsymbol{r}, t)$), is constant in body rest frame
−Explicit expression for components:

$$I_{jk} = \int dV\, \rho(\boldsymbol{r},\, t) \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix}$$

*Diagonal components (red) coincide with moment of inertia around axes x, y, z respectively:

$$I_{xx} \leftrightarrow I_x, \qquad I_{yy} \leftrightarrow I_y, \qquad I_{zz} \leftrightarrow I_z$$

Properties of tensor of inertia

$$I_{jk} = \int d^3r\, \rho(\boldsymbol{r}, t) \left(\delta_{jk}\mathfrak{r}^2 - \mathfrak{r}_j \mathfrak{r}_k\right)$$

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*Diagonal components (red) coincide with moment of inertia around axes x, y, z respectively:

$$I_{xx} \quad \leftrightarrow I_x, \qquad I_{yy} \leftrightarrow I_y, \qquad I_{zz} \leftrightarrow I_z$$

Lemma

For rotation around any axis $\boldsymbol{n}$, its moment of inertia I_n is given by

$$I_n = \sum_{jk} I_{jk} n_j n_k = \vec{n} \cdot \hat{I} \cdot \vec{n} \qquad (1)$$

e.g. for rotation around axis z we have $\boldsymbol{n} = (0, 0, 1)$, so $I_z = I_{zz}$

Proof:

- Kinetic energy of rotation:

$$T_{\mathrm{rot}} = \frac{1}{2} \int dV \, \rho\left(\vec{r}\right) \left[\boldsymbol{\Omega} \times \boldsymbol{r}\right]^2$$

Let's decompose vector $\boldsymbol{r}$ into parts $||$ and $\perp$ to $\boldsymbol{\Omega}$,

$$\boldsymbol{r} \quad = \boldsymbol{r}_{||} + \boldsymbol{r}_{\perp}, \qquad \Omega \cdot r_{\perp} = 0, \, \boldsymbol{\Omega} \times \boldsymbol{r}_{||} = 0$$

and introduce unit vector $\boldsymbol{n} = \boldsymbol{\Omega}/|\Omega|$, so $\boldsymbol{\Omega} = \Omega \boldsymbol{n}$

$$r_{||} = \boldsymbol{n}\left(n \cdot r\right), \quad \boldsymbol{r}_{\perp} = \boldsymbol{r} - \boldsymbol{r}_{||} = \boldsymbol{r} - \boldsymbol{n}\left(n \cdot r\right)$$

$$\Rightarrow T_{\mathrm{rot}} = \frac{1}{2} \int dV \, \rho\left(\vec{r}\right) \left[\boldsymbol{\Omega} \times \boldsymbol{r}_{\perp}\right]^2 = \frac{1}{2} \Omega^2 \underbrace{\int dV \, \rho\left(\vec{r}\right) \boldsymbol{r}_{\perp}^2}_{:= I_n} = \frac{I_{jk} \Omega_j \Omega_k}{2} = \frac{I_{jk} n_j n_k}{2} \Omega^2$$

From comparison of magenta- and red-colored expressions conclude that (1) is valid, Q.E.D.

Tensor of inertia: transformation of components

$$I_{jk} = \int dV\, \rho(r) \left(\delta_{jk} r^2 - r_j r_k\right) \qquad (1)$$

- Identify how components of tensor of inertia transform under rotations of coordinates

$$r_i \to r_i = \sum_{i'} \mathcal{R}_{ii'} r_{i'}$$

As You may remember, in case of rotations $\mathcal{R}_{ij}$ are orthogonal matrices, e.g:

$$\hat{\mathcal{R}}_Z = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix} = \hat{\mathcal{R}}_Z \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x\cos\theta + y\sin\theta \\ -x\sin\theta + y\cos\theta \\ z \end{pmatrix}$$

The idea is to express I_{ij} in different frames (e.g. lab frame and body's rest frame) to be able to relate them (and thus simplify analysis)

Tensor of inertia: transformation of components

$$I_{jk} = \int dV\,\rho(r)\left(\delta_{jk} r^2 - r_j r_k\right) \qquad (1)$$

– From definition (1): second term gets two $\hat{\mathcal{R}}$-matrices, one for each "free" index

$$r_j r_k \to \sum_{j'k'} \mathcal{R}_{jj'}\mathcal{R}_{kk'} r_{j'} r_{k'}$$

so it "looks like" that I_{jk} should transform similarly as

$$I_{jk} \to \sum_{j'k'} \mathcal{R}_{jj'}\mathcal{R}_{kk'} I_{j'k'} \qquad (2)$$

–The term $\sim r^2\delta_{ik}$ is invariant (r^2 is absolute value of radius-vector and does not change).

⇒Seems we are in trouble, two terms in (1) transform differently (?)

… but we recall that

$$\sum_{j'k'} \mathcal{R}_{j\,j'}\mathcal{R}_{k\,k'}\delta_{j'k'} = \sum_{j'} \mathcal{R}_{j\,j'}\mathcal{R}_{k\,j'} = \delta_{jk}$$

(orthogonality), so can interpret that it also acquires two $\hat{\mathcal{R}}$-matrices per each index (!)

⇒ The tensor I_{jk} transforms as given in (2).

Tensor of inertia: transformation of components

$$I_{jk} \to \sum_{j'k'} \mathcal{R}_{jj'} \mathcal{R}_{kk'} I_{j'k'} \qquad (1)$$

- This result is very important:

–In any frame

$$I_{jk} = \int dV\, \rho(\mathbf{r},\ t) \left(\delta_{jk} r^2 - r_j r_k \right) =$$
$$= \int dV\, \rho(\mathbf{r},\ t) \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix}$$

– in lab frame the density depends on orientation of the body (=on time), for this reason all components $I_{ij}^{(\mathrm{lab})}$ also depend on time

–in body's rest frame ρ does not depend on time, so $I_{ij}^{(\mathrm{body})}$ are constant

–The equation (1) allows to relate $I_{ij}^{(\mathrm{lab})}$ and $I_{ij}^{(\mathrm{body})}$ if we know orientation of the body (and associated rotation matrices $\mathcal{R}$)

Tensor of inertia - Example

$$I_{jk} = \int dV\, \rho(r) \left(\delta_{jk} r^2 - r_j r_k\right)$$

Evaluate the tensor of inertia for the homogeneous sphere of radius R and mass m.

- The moments of inertia $I_x = I_y = I_z = \frac{2}{5} m R^2$, which by definition coincide with components I_{xx}, I_{yy}, I_{zz}
- For nondiagonal components we have:

$$I_{xy} = -\int \rho dV\, x\, y = -\rho \int_0^R r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi\, r^2 \sin^2\theta \underbrace{\sin\phi\cos\phi}_{\frac{1}{2}\sin 2\phi} = 0$$

$$I_{xz} = -\int \rho dV\, x\, z = -\rho \int_0^R r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi\, r^2 \frac{\sin 2\theta}{2} \cos\phi = 0$$

$$I_{yz} = -\int \rho dV\, y\, z = -\rho \int_0^R r^2 dr \int_0^\pi \sin\theta d\theta \int_0^{2\pi} d\phi\, r^2 \frac{\sin 2\theta}{2} \sin\phi = 0$$

Tensor of inertia - Example

$$\Rightarrow I_{jk} = \frac{2}{5} mR^2 \delta_{jk} = \frac{2}{5} mR^2 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

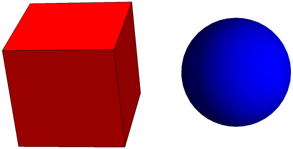
Why nondiagonal components vanish?

The sphere has 3 symmetry planes (invariant w.r.t. 3 *independent* transformations $x \to -x$, $y \to -y$, $z \to -z$ in Cartesian coordinates), and nondiagonal components change sign under such transforms, that's why they vanished

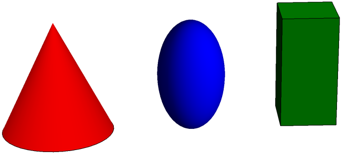
Tensor of inertia - Classification

►Since I_{jk} is symmetric, real-valued, can always diagonalize it
−Physically three different scenarios:

- $I_1 = I_2 = I_3$-spherical top (spherically symmetric)



- $I_1 = I_2 \neq I_3$-symmetric top (axially symmetric)



- $I_1 \neq I_2 \neq I_3$-asymmetric top

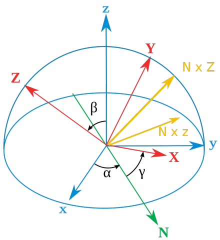
−whatever object without symmetries

Description in lagrangian formalism

- Now we'll try to formulate and solve the problem of motion of a rigid body in lagrangian formalism
  - First we'll introduce Euler's angles which describe finite rotations
  - We'll work in the body's rest frame, since $I_{ij} = \mathrm{const}$ there
  - Next we'll construct the Lagrangian and try to solve equations of motion (using possible integrals of motion)

Angular velocity

Assume that the orientation of rigid body is described in terms of Euler angles α, β, γ. Evaluate the components of vector $\vec{\Omega}$ in the rest frame of the body



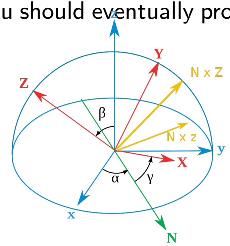
Description in lagrangian formalism

- It is convenient to decompose the angular vector $\boldsymbol{\Omega}$ as

$$\boldsymbol{\Omega} = \dot{\alpha}\boldsymbol{e}_\alpha + \dot{\beta}\boldsymbol{e}_\beta + \dot{\gamma}\boldsymbol{e}_\gamma$$

where $\boldsymbol{e}_\alpha$, $\boldsymbol{e}_\beta$, $\boldsymbol{e}_\gamma$ are unit vectors in directions of corresponding rotation axes:
▷ $\boldsymbol{e}_\alpha$ points in vertical direction of old axis Z (blue arrow)
▷$\boldsymbol{e}_\beta$ points in direction of line of nodes N (green arrow)
▷$\boldsymbol{e}_\gamma$ points in direction of line of new axis Z (red arrow)

- You should eventually project vector $\boldsymbol{\Omega}$ onto the axes X, Y, Z of the body rest frame

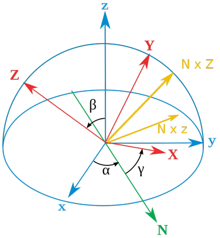


Vector $\boldsymbol{\Omega}$ in the body rest frame:

$$\Omega_1 = \dot{\alpha}\sin\beta\sin\gamma + \dot{\beta}\cos\gamma$$
$$\Omega_2 = \dot{\alpha}\sin\beta\cos\gamma - \dot{\beta}\sin\gamma$$
$$\Omega_3 = \dot{\alpha}\cos\beta + \dot{\gamma}.$$

(can get the result making projections geometrically)

Description in lagrangian formalism



Vector $\vec{\Omega}$ in the body rest frame:

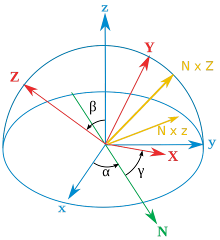
$$\Omega_1 = \dot{\alpha} \sin\beta \sin\gamma + \dot{\beta} \cos\gamma$$
$$\Omega_2 = \dot{\alpha} \sin\beta \cos\gamma - \dot{\beta} \sin\gamma$$
$$\Omega_3 = \dot{\alpha} \cos\beta + \dot{\gamma}.$$

Kinetic energy

$$T = \frac{I_1 \Omega_1^2}{2} + \frac{I_2 \Omega_2^2}{2} + \frac{I_3 \Omega_3^2}{2}$$

−direct substitution gives result (lengthy but simple by structure)

Description in lagrangian formalism



Vector $\vec{\Omega}$ in the body rest frame:

$$\Omega_1 = \dot{\alpha} \sin\beta \sin\gamma + \dot{\beta} \cos\gamma$$
$$\Omega_2 = \dot{\alpha} \sin\beta \cos\gamma - \dot{\beta} \sin\gamma$$
$$\Omega_3 = \dot{\alpha} \cos\beta + \dot{\gamma}.$$

Symmetric top

Write out the kinetic energy of the symmetric top ($I_1 = I_2 \neq I_3$) whose orientation is described by Euler angles α, β, γ. Write out the Lagrangian for the case of free motion and integrate the EOM.

Description in lagrangian formalism

Symmetric top:

►Kinetic energy and lagangian:

$$\begin{aligned} T &= \frac{I_1}{2}\left(\dot{\alpha}^2 \sin^2\beta + \dot{\beta}^2\right) + \frac{I_3}{2}\left(\dot{\alpha}\cos\beta + \dot{\gamma}\right)^2 \\ U &= 0 \Rightarrow T = E = L = \mathrm{const} \end{aligned}$$

Generalized momenta ($p_\alpha = \partial L/\partial \dot{q}_a$):

$$\begin{aligned} M_\beta &= I_1\dot{\beta} \neq \mathrm{const}\,(\beta\ \ \mathrm{not\ cyclical}) \\ M_\alpha &= \dot{\alpha}\left(I_1 \sin^2\beta + I_3\cos^2\beta\right) + I_3\dot{\gamma}\cos\beta = \mathrm{const} \\ M_\gamma &= I_3\left(\dot{\alpha}\cos\beta + \dot{\gamma}\right) = \mathrm{const}. \end{aligned}$$

►Note that $\boldsymbol{M} = \mathrm{const}$ and in lab frame all 3 components M_x, M_y, M_x are conserved.

► In moving frame directions of basis vectors $\boldsymbol{e}_\alpha$, $\boldsymbol{e}_\beta$, $\boldsymbol{e}_\gamma$ change, so projections of $\boldsymbol{M}$ onto these axes are NOT constants

$$\dot{\alpha} = \frac{M_\alpha - M_\gamma\cos\beta}{\tilde{I}_1\sin^2\beta}, \quad \dot{\gamma} = \frac{M_\gamma}{I_3} - \cos\beta\frac{M_\alpha - M_\gamma\cos\beta}{\tilde{I}_1\sin^2\beta}.$$

Description in lagrangian formalism

$$\dot\alpha = \frac{M_\alpha - M_\gamma \cos\beta}{\tilde I_1 \sin^2\beta}, \quad \dot\gamma = \frac{M_\gamma}{I_3} - \cos\beta \frac{M_\alpha - M_\gamma\cos\beta}{\tilde I_1 \sin^2\beta}.$$

$$\Rightarrow E = \frac{M_\gamma^2}{2I_3} + \frac{I_1}{2}\dot\beta^2 + \frac{(M_\alpha - M_\gamma\cos\beta)^2}{2I_1\sin^2\beta}.$$

$$\lambda = \cos\beta; \quad \dot\lambda = -\dot\beta\sin\beta,$$

$$\dot\beta = -\frac{\lambda}{\sqrt{1-\lambda^2}}.$$

$$\frac{I_1}{2}\dot\lambda^2 + \lambda^2 \underbrace{\left(E + \frac{M_\gamma^2}{2}\left(\frac{1}{I_1} - \frac{1}{I_3}\right)\right)}_{a/2} -$$

$$-\underbrace{\frac{M_\alpha M_\gamma}{I_1}}_{b}\lambda + \underbrace{\left(\frac{M_\alpha^2}{2I_1} + \frac{M_\gamma^2}{2I_3} - E\right)}_{c/2} = 0$$

Description in lagrangian formalism

$$\Rightarrow E = \frac{M_\gamma^2}{2I_3} + \frac{I_1}{2}\dot{\beta}^2 + \frac{(M_\alpha - M_\gamma \cos\beta)^2}{2I_1 \sin^2\beta}.$$

$$\lambda \quad = \cos\beta,$$

$$\frac{I_1}{2}\dot{\lambda}^2 + \lambda^2\frac{a}{2} - b\lambda + \frac{c}{2} = 0$$

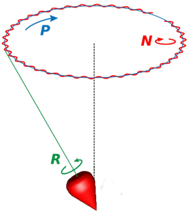
►Introduce $\mu(t) := \lambda(t) - \frac{b}{a}$, can get

$$\frac{I_1}{2}\dot{\mu}^2 + \frac{a}{2}\mu^2 = \underbrace{\frac{b^2}{2a} - \frac{c}{2}}_{=\mathcal{E}}$$

$\Rightarrow$coincides with expression for harmonic oscillator with "mass" I_1, "elasticity" a and energy $\mathcal{E}$

$$\Rightarrow \qquad \mu(t) = A\cos(\omega t + \varphi_0),$$
$$A = \sqrt{2\mathcal{E}/a}, \quad \omega = \sqrt{a/I_1}$$

$$\Rightarrow \cos\beta = \frac{b}{a} + A\cos(\omega t + \varphi_0)$$



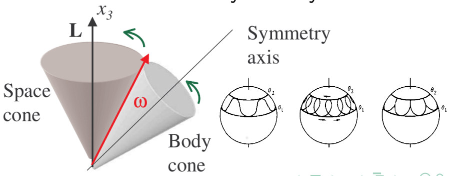
oscillations with frequency

$$\omega^2 = \frac{a}{I_1},$$

around "center of equilibrium"

$$\lambda_0 \equiv \cos\beta_0 = \frac{b}{a},$$

- Visualization of the symmetry axis motion



Summary

Main results:

−Concept of the differential cross-section

−Cross-section in the central field.

  *Rutherford's formula (cross-section in the Coulomb field)

  *Capture cross-section

−Inverse problem of scattering. Focusing potentials